

Introduction to Python for Quantum Computing

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1. Why Python, and how this notebook works
2. Python fundamentals
3. Data structures
4. Functions and a first look at classes/objects
5. Packages and environments
6. Scientific Python: NumPy, SciPy, Matplotlib
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Introduction to Python for Quantum Computing

Quantum Computing Spring School 2026

Session 1 · ~90 minutes

Welcome! This notebook is a general, hands-on introduction to Python and its core scientific libraries. No prior programming experience is assumed.

What we'll cover today:

1. Why Python, and how Jupyter notebooks work
2. Python fundamentals — variables, types, operators, control flow
3. Data structures — lists, tuples, dictionaries, sets
4. Functions and a first look at classes/objects
5. Packages and environments — Anaconda, `conda`, `pip`
6. Scientific Python — NumPy, SciPy, Matplotlib

This session is deliberately **general-purpose Python** — it does not touch quantum computing concepts, qubits, or Qiskit. That's the focus of a later session, once you have the Python fluency to focus entirely on the quantum ideas rather than fighting the language at the same time.

Each section has one or more **Exercises** — do these yourself before looking at the solution cell underneath.

1. Why Python, and how this notebook works

Python is one of the most widely used programming languages in research and scientific computing — it's readable, has a huge ecosystem of libraries (including NumPy, SciPy, and Matplotlib, which we'll use today), and doesn't require you to manage memory or compile anything: you write code and run it immediately. You'll keep building on today's skills throughout the rest of the school.

This document is a **Jupyter notebook**. It's made of *cells*:

- ♦ **Markdown cells** (like this one) hold formatted text, explanations, and headings.
- ♦ **Code cells** hold Python code you can run.

To run a code cell in Jupyter: click it and press **Shift+Enter** (runs it, moves to the next cell) or **Ctrl+Enter** (runs it, stays put). Try it on the cell below.

```
print("Hello, Spring School!")
```

A few notebook habits worth knowing immediately:

- ♦ Cells share state — a variable defined in one cell is available in later cells, in the order you *ran* them (not necessarily the order they appear on the page). Restart-and-run-all before you trust a notebook end to end.
- ♦ The output of the **last line** of a cell is displayed automatically, without needing `print`:

```
3 + 4 # this shows "7" as output, even without print()
```

- ♦ `print()` is still needed if you want to show something that *isn't* the last line, or want multiple outputs from one cell.

```
print("first line")
print("second line")
5 * 5 # only this last expression's value is auto-displayed, after both prints
```

2. Python fundamentals

2.1 Variables and assignment

A variable is a name pointing at a value. Python figures out the type for you — you don't declare it up front (this is called *dynamic typing*).

```
book_title = "Dune"
price = 24.99
in_stock = True

print(book_title, price, in_stock)
```

You can reassign a variable to a different type entirely — Python won't stop you (whether that's wise is a separate question):

```
x = 5
print(type(x))

x = "now I'm text"
print(type(x))
```

2.2 Basic types

The core built-in types you'll use constantly:

```
an_int = 42 # int
a_float = 3.14159 # float
a_complex = 2 + 3j # complex — note the `j`, not `i`
a_string = "hello" # str
a_bool = True # bool

for value in [an_int, a_float, a_complex, a_string, a_bool]:
    print(value, "->", type(value))
```

Python has native support for complex numbers, which is worth knowing even outside any specific application:

```
z = 2 + 3j
print("real part:", z.real)
print("imaginary part:", z.imag)
print("magnitude:", abs(z))
print("conjugate:", z.conjugate())
```

Exercise: create two variables `a = 3` and `b = 4.5`, then print their sum, their product, and the type of each result. (What type do you get when you add an `int` and a `float`?)

```
a = 3
b = 4.5
print("sum:", a + b, type(a + b))
print("product:", a * b, type(a * b))
```

2.3 Operators

Arithmetic works about how you'd expect, with a couple of Python-specific ones:

```
a, b = 7, 2

print("a + b =", a + b)
print("a - b =", a - b)
print("a * b =", a * b)
print("a / b =", a / b)      # true division -> always a float
print("a // b =", a // b)   # floor division -> integer result
print("a % b =", a % b)     # remainder / modulo
print("a ** b =", a ** b)   # exponentiation
```

Comparisons return booleans, and can be combined with `and`, `or`, `not`:

```
print(a > b)
print(a == 7)
print(a > b and b > 0)
print(not in_stock)
```

Exercise: given an integer `n`, use `%` (modulo) and an `if/else` to print whether `n` is even or odd. Test it on `n = 17` and `n = 42`.

```
def even_or_odd(n):
    if n % 2 == 0:
        return "even"
    else:
        return "odd"

for n in [17, 42]:
    print(n, "is", even_or_odd(n))
```

2.4 Strings

Strings support slicing, concatenation, and a large set of built-in methods.

```
name = "Ada Lovelace"

print(name.lower())
print(name.upper())
print(name.replace("Ada", "Grace"))
print(len(name))
print(name[0])      # first character
print(name[-1])     # last character
print(name[0:3])    # slice: characters 0,1,2
```

f-strings (`f"..."`) are the modern way to build strings with variables embedded in them — you'll use these constantly for labelling output and plots:

```
event = "the Quantum Computing Spring School"
n_days = 3
print(f"Welcome to {event} – {n_days} days of sessions ahead")
print(f"That's {n_days * 24} hours, roughly")
```

Exercise: given `full_name = "Grace Hopper"`, print just the first name and just the last name using slicing or string methods (hint: `.split()` splits a string into a list of words).

```
full_name = "Grace Hopper"
first, last = full_name.split()
print("first name:", first)
print("last name:", last)
```

2.5 Control flow: `if` / `elif` / `else`

```
score = 72

if score >= 90:
    grade = "A"
elif score >= 70:
    grade = "B"
elif score >= 50:
    grade = "C"
else:
    grade = "F"

print(f"Score {score} -> grade {grade}")
```

Exercise: write an `if/elif/else` block that classifies a temperature (in °C) as "freezing" (≤ 0), "cold" (0–15), "mild" (15–25), or "hot" (> 25). Test it with a few different values.

```
def classify_temperature(temperature):
    if temperature <= 0:
        return "freezing"
    elif temperature <= 15:
        return "cold"
    elif temperature <= 25:
        return "mild"
    else:
        return "hot"

for t in [-5, 10, 20, 30]:
    print(t, "->", classify_temperature(t))
```

2.6 Loops: `for` and `while`

`for` loops iterate over a sequence (a list, a range of numbers, characters in a string, ...):

```
for i in range(5):          # 0, 1, 2, 3, 4 – stop is exclusive
    print("iteration", i)
```

```
titles = ["Dune", "Foundation", "Neuromancer"]
for title in titles:
    print("shelving:", title)
```

`while` loops run until a condition becomes false — useful when you don't know the number of iterations in advance:

```
balance = 100.0
years = 0
while balance < 200.0:
    balance *= 1.05    # 5% growth per "year"
```

```
years += 1

print(f"Took {years} years to double, ending at {balance:.2f}")
```

Exercise: use a `for` loop to compute the sum of the squares of the numbers 1 to 10 (i.e. $1^2 + 2^2 + \dots + 10^2$).

```
total = 0
for i in range(1, 11):
    total += i ** 2

print(total)
```

3. Data structures

3.1 Lists — ordered, mutable

```
grades = [72, 88, 91, 64, 79]
print(grades)
print(grades[0])      # indexing
print(grades[-1])     # last element
print(grades[1:3])    # slicing

grades.append(85)      # add to the end
print(grades)

grades.remove(64)      # remove by value
print(grades)

print(len(grades))
print(sum(grades) / len(grades))  # average
```

Lists can hold mixed types, and can be nested:

```
student = ["Alex", 21, 88.5]  # name, age, average grade
print(student)

class_lists = ["Alex", "Sam"], ["Priya", "Jordan"], ["Lee"]  # groups of students
print(class_lists[1])      # second group
print(class_lists[1][0])   # first student in the second group
```

Exercise: given a list of prices, use a loop (or a list comprehension, if you're comfortable with section 3.5) to build a new list containing only the prices greater than R100, then print the total of that filtered list.

```
prices = [45.00, 120.50, 89.99, 210.00, 65.25, 150.00]

expensive = [p for p in prices if p > 100]
print("expensive items:", expensive)
print("total:", sum(expensive))
```

3.2 Tuples — ordered, immutable

Tuples look like lists but can't be changed after creation. Use them for fixed, small groupings — e.g. coordinates, or a (name, value) pair.

```

coordinate = (3, 4)    # (x, y)
print(coordinate)
print(coordinate[0])

x, y = coordinate    # unpacking
print(f"x={x}, y={y}")

# coordinate[0] = 10    # uncommenting this raises a TypeError – tuples are immutable

```

A common pattern: a list of tuples, unpacked directly in a `for` loop.

```

scores = [("Alex", 72), ("Sam", 88), ("Priya", 91), ("Jordan", 64)]

for name, score in scores:
    print(f"{name}: {score}")

```

Exercise: using the `scores` list above, find and print the name of the student with the *highest* score, without using `max()` directly on the tuples (hint: loop through, and keep track of the best name/score seen so far).

```

best_name, best_score = scores[0]
for name, score in scores[1:]:
    if score > best_score:
        best_name, best_score = name, score

print(f"Top score: {best_name} with {best_score}")

```

3.3 Dictionaries — key/value pairs

```

book_stock = {"Dune": 12, "Foundation": 5, "Neuromancer": 0}    # title -> copies in stock

print(book_stock["Dune"])
print(book_stock.keys())
print(book_stock.values())
print(book_stock.items())

for title, copies in book_stock.items():
    print(f"{title}: {copies} in stock")

```

```

out_of_stock = [title for title, copies in book_stock.items() if copies == 0]
print(out_of_stock)

```

3.4 Sets — unordered, unique elements

```

genres_read = {"sci-fi", "fantasy", "sci-fi", "mystery", "fantasy"}    # duplicates are dropped
↳ automatically
print(genres_read)
print(len(genres_read))

all_genres = {"sci-fi", "fantasy", "mystery", "romance", "horror"}
print(all_genres - genres_read)    # genres never read

```

Exercise: you're given a list of words. Build a dictionary mapping each unique word to how many times it appears (a simple word-count / frequency table).

```
words = ["the", "quick", "brown", "fox", "the", "fox", "jumps", "the"]

word_counts = {}
for word in words:
    word_counts[word] = word_counts.get(word, 0) + 1

print(word_counts)
```

3.5 Comprehensions (stretch)

A compact way to build a list/dict/set from a loop in one line. Optional, but you'll see this style constantly in other people's code.

```
# Long form
squares = []
for i in range(6):
    squares.append(i ** 2)
print(squares)

# Equivalent list comprehension
squares = [i ** 2 for i in range(6)]
print(squares)

# With a filter condition
even_squares = [i ** 2 for i in range(10) if i % 2 == 0]
print(even_squares)

# Dict comprehension
prices = {"Dune": 24.99, "Foundation": 19.50, "Neuromancer": 15.00}
discounted = {title: round(price * 0.9, 2) for title, price in prices.items()}
print(discounted)
```

Exercise: using a list comprehension, build a list of the *cubes* (³) of every odd number from 1 to 9.

```
cubes_of_odds = [n ** 3 for n in range(1, 10) if n % 2 != 0]
print(cubes_of_odds)
```

4. Functions and a first look at classes/objects

4.1 Functions

A function packages up reusable logic. `def` defines it; `return` sends back a value.

```
def average(numbers):
    """Return the arithmetic mean of a list of numbers."""
    return sum(numbers) / len(numbers)

print(average([72, 88, 91, 64, 79]))
```

Default arguments and keyword arguments:

```
def describe_book(title, pages=100, genre="unknown"):
    print(f'{title} - {pages} pages - genre: {genre}')

describe_book("Dune") # uses the defaults
describe_book("Dune", pages=412) # override one default
```

```
describe_book(title="Foundation", genre="sci-fi")           # named arguments, any order
```

Exercise: write a function `celsius_to_fahrenheit(c)` that converts a Celsius temperature to Fahrenheit ($F = C * 9/5 + 32$), then use it to print the Fahrenheit equivalents of `[0, 20, 37, 100]`.

```
def celsius_to_fahrenheit(c):
    return c * 9 / 5 + 32

for c in [0, 20, 37, 100]:
    print(f"{c}°C -> {celsius_to_fahrenheit(c)}°F")
```

4.2 Classes and objects

Everything in Python is an object, and most non-trivial libraries you'll use are built from **classes** — recognising the pattern will make library documentation far less mysterious, whatever library it happens to be.

A **class** is a blueprint; an **object** (or *instance*) is a specific thing built from that blueprint. You've actually been using objects since the start of this notebook without naming them as such — a string is an object, and `.lower()`, `.upper()`, `.split()` are all **methods** being called on it; a list is an object too, and `.append()`, `.remove()` are its methods.

```
name = "Ada Lovelace"
print(name.lower())    # calling a method on a string object

grades = [72, 88, 91]
grades.append(85)      # calling a method on a list object
print(grades)
```

The rest of this section is about building *your own* objects — the exact same idea, just with a class you define yourself.

```
class Book:
    """A simple representation of a book."""

    def __init__(self, title, author, price):
        # __init__ runs when you create a new Book – it sets up the object's state
        self.title = title
        self.author = author
        self.price = price

    def discounted_price(self, percent_off):
        # a method: a function that belongs to the class and acts on `self`
        return self.price * (1 - percent_off / 100)

    def __repr__(self):
        # controls how the object prints
        return f"Book('{self.title}' by {self.author}, R{self.price:.2f})"

b1 = Book("Dune", "Frank Herbert", 249.99)
b2 = Book("Foundation", "Isaac Asimov", 199.50)

print(b1)
print(b2)
print(b1.discounted_price(20))
print(b1.title, "vs", b2.title)
```

Key vocabulary you'll meet again and again in other people's code:

- ♦ **self** refers to the specific object a method is being called on.
- ♦ **__init__** is the constructor — it runs automatically when you write `Book(...)`.
- ♦ An **attribute** (`b1.title`) is data stored on the object.
- ♦ A **method** (`b1.discounted_price(20)`) is a function stored on the object.

```
library = [Book("Dune", "Frank Herbert", 249.99),
           Book("Foundation", "Isaac Asimov", 199.50),
           Book("Neuromancer", "William Gibson", 179.00)]

for book in library:
    print(book, "-> 10% off:", round(book.discounted_price(10), 2))
```

Exercise: add a method `is_expensive(self, threshold=200)` to the `Book` class above that returns `True` if `self.price` is greater than `threshold`. Test it on `b1` and `b2`.

```
class Book:
    def __init__(self, title, author, price):
        self.title = title
        self.author = author
        self.price = price

    def discounted_price(self, percent_off):
        return self.price * (1 - percent_off / 100)

    def is_expensive(self, threshold=200):
        return self.price > threshold

    def __repr__(self):
        return f"Book('{self.title}' by {self.author}, R{self.price:.2f})"

b1 = Book("Dune", "Frank Herbert", 249.99)
b2 = Book("Foundation", "Isaac Asimov", 199.50)

print(b1.title, "expensive?", b1.is_expensive())
print(b2.title, "expensive?", b2.is_expensive())
```

Objects containing objects

Real libraries rarely stop at one class — bigger, more useful objects are usually built by **combining** smaller ones (this is called *composition*). Here, a `ShoppingCart` object holds a list of `Item` objects, and its methods operate on that whole collection:

```
class Item:
    def __init__(self, name, price, quantity=1):
        self.name = name
        self.price = price
        self.quantity = quantity

    def subtotal(self):
        return self.price * self.quantity

    def __repr__(self):
        return f"Item({self.name}, R{self.price:.2f} x{self.quantity})"
```

```

class ShoppingCart:
    def __init__(self, owner):
        self.owner = owner
        self.items = []          # a ShoppingCart *contains* Item objects

    def add_item(self, item):
        self.items.append(item)

    def total(self):
        return sum(item.subtotal() for item in self.items)

    def __repr__(self):
        return f"ShoppingCart({self.owner}, {len(self.items)} items, total=R{self.total():.2f})"

cart = ShoppingCart("Ian")
cart.add_item(Item("Notebook", 45.00, quantity=3))
cart.add_item(Item("Pen", 12.50, quantity=5))

print(cart)
for item in cart.items:
    print(" -", item, "-> subtotal:", item.subtotal())
print("Grand total:", cart.total())

```

Notice `cart.items` is a list of `Item` objects, not raw numbers — `cart.total()` has to reach into each one and call `.subtotal()` on it. This “objects holding other objects, with methods that operate across all of them” pattern shows up everywhere once you start using real packages — a Matplotlib `Figure` holds `Axes` objects, a pandas `DataFrame` holds `Series` objects, and so on.

Objects you didn’t build yourself

Every object from a package you `import` follows exactly the same rules as `Book` and `ShoppingCart` above — a class defines it, you create an instance, and you call methods/read attributes on that instance. Two everyday examples from Python’s standard library:

```

from datetime import date

today = date.today()          # creating an object from the datetime package's Date class
print(today)                  # __repr__ controls how it prints, just like our classes
print(today.year, today.month, today.day) # attributes
print(today.strftime("%A, %d %B %Y"))    # a method call, returning a formatted string

```

```

from pathlib import Path

notebook_folder = Path("session-1-python-intro") # a Path object
print(notebook_folder)
print(notebook_folder.name)                     # an attribute
print(notebook_folder.suffix)                   # another attribute (empty here – it's a folder, not a file)
print(notebook_folder / "01_intro_python_for_quantum_computing.ipynb") # '/' is overloaded as a
↳ method here!

```

Recognising “this is an object, that’s a method call, that’s an attribute” is most of what you need to read unfamiliar package documentation — including the scientific packages in the next section, and whatever tools you meet later in the school.

Exercise: write a `Student` class with attributes `name` and `grades` (a list, empty by default), a

method `add_grade(self, grade)` that appends to `self.grades`, and a method `average(self)` that returns the mean of `self.grades` (return `0` if there are no grades yet, to avoid dividing by zero). Create one `Student`, add three grades, and print their average.

```
class Student:
    def __init__(self, name):
        self.name = name
        self.grades = []

    def add_grade(self, grade):
        self.grades.append(grade)

    def average(self):
        if not self.grades:
            return 0
        return sum(self.grades) / len(self.grades)

    def __repr__(self):
        return f"Student({self.name}, avg={self.average():.1f})"

s1 = Student("Priya")
s1.add_grade(78)
s1.add_grade(85)
s1.add_grade(91)

print(s1)
print(s1.name, "average:", s1.average())
```

5. Packages and environments

So far we've used only Python's *built-ins*. Real projects rely on external **packages** — pre-written code you install and `import`. NumPy, SciPy and Matplotlib, which we'll use in the next section, are all packages.

5.1 Anaconda and environments

Anaconda is a Python distribution: it bundles Python itself with `conda` (a package/environment manager) and a large set of scientific packages pre-installed.

A **virtual environment** is an isolated, self-contained set of packages for one project. Why bother?

- ♦ Different projects can need different (even conflicting) package versions.
- ♦ You can experiment freely without breaking your main Python install.
- ♦ It makes your work reproducible — someone else can recreate exactly your setup.

Common `conda` commands (run these in a terminal / Anaconda Prompt, **not** in this notebook):

```
# create a new environment named "saqutischool" with Python 3.14
conda create -n saqutischool python=3.14

# activate it (do this every time you start work)
conda activate saqutischool

# see all environments you have
conda env list
```

```
# install a package into the active environment – no version pinned, so this
# always grabs the latest release available for Python 3.14
conda install numpy scipy matplotlib

# leave the environment
conda deactivate
```

Keeping a dedicated environment for this school means today's setup can't silently interfere with — or be broken by — anything else on your laptop. We're pinning the **Python version** (3.14) so everyone's environment behaves the same way, but deliberately *not* pinning package versions — every install above pulls whatever is currently the latest release.

5.2 pip and installing packages

`conda install` covers most scientific packages, but many packages are installed with **pip**, Python's own package installer, which pulls from PyPI (the Python Package Index). Both can be used inside the same conda environment.

```
# with your environment activated:
pip install requests

# see what's installed
pip list

# check a specific package's version
pip show numpy
```

(A full, step-by-step setup sequence — including the packages needed for later sessions — is in the pre-school setup guide ([setup/pre_school_setup_guide.pdf](#)). Please do this before you arrive!)

5.3 import, back inside Python

Once a package is installed, `import` brings it into your notebook. This is the one part of this section you *do* run here:

```
import math

print(math.pi)
print(math.sqrt(2))
print(math.sin(math.pi / 2))
```

You can import just part of a package, or give it a short alias — you'll see `import numpy as np` in almost every scientific Python notebook you ever open:

```
from math import pi, sqrt
print(pi, sqrt(4))

import math as m
print(m.e)
```

Exercise: `import random` and use `random.randint(1, 6)` to simulate rolling a six-sided die 10 times, storing each result in a list. Then print the list and its average.

```
import random

rolls = [random.randint(1, 6) for _ in range(10)]
```

```
print(rolls)
print("average:", sum(rolls) / len(rolls))
```

6. Scientific Python: NumPy, SciPy, Matplotlib

These three packages are the backbone of numerical work in Python. We'll work through a small running example — a week of temperature readings from a weather station.

6.1 NumPy — arrays and linear algebra

NumPy's core object is the **array** — a grid of numbers (usually all the same type) that supports fast, vectorised math.

```
import numpy as np

temperatures = np.array([18.5, 19.0, 21.2, 20.8, 17.9, 16.4, 19.3]) # one week, °C
print(temperatures)
print(type(temperatures))
print(temperatures.shape)    # (7,) -> 1-dimensional, 7 entries
print(temperatures.dtype)    # the data type NumPy chose
```

Vectorised arithmetic — operations apply element-wise, without writing a loop:

```
fahrenheit = temperatures * 9 / 5 + 32
print(fahrenheit)

print("mean:", temperatures.mean())
print("std dev:", temperatures.std())
print("min/max:", temperatures.min(), temperatures.max())
```

Comparing plain Python lists to NumPy arrays — a common early trip-up:

```
python_list = [1, 2, 3]
numpy_array = np.array([1, 2, 3])

print(python_list + python_list)    # concatenates: [1, 2, 3, 1, 2, 3]
print(numpy_array + numpy_array)    # adds element-wise: [2, 4, 6]
```

2D arrays (matrices) and matrix operations:

```
matrix = np.array([[1, 2, 3],
                   [4, 5, 6],
                   [7, 8, 9]])

print(matrix)
print(matrix.shape)
print(matrix[0, :])    # first row
print(matrix[:, 1])    # second column
print(matrix.T)        # transpose

vector = np.array([1, 0, -1])
print(matrix @ vector)  # matrix-vector multiplication with @
```

Other array-building and indexing tools you'll use constantly:

```
print(np.zeros(4))
print(np.ones(4))
```

```
print(np.linspace(0, 1, 5))    # 5 evenly spaced points from 0 to 1
print(np.arange(0, 10, 2))    # like range(), but returns an array
```

Exercise: create a NumPy array `heights` with values [1.65, 1.72, 1.58, 1.80, 1.69] (metres). Compute and print the mean height, the height of the tallest person, and each person's height converted to centimetres.

```
heights = np.array([1.65, 1.72, 1.58, 1.80, 1.69])
print("mean height (m):", heights.mean())
print("tallest (m):", heights.max())
print("heights (cm):", heights * 100)
```

6.2 SciPy — a layer above NumPy

SciPy builds on NumPy with more specialised numerical routines: optimisation, integration, statistics, and linear algebra beyond what `numpy.linalg` covers.

Eigenvalues/eigenvectors of a matrix:

```
from scipy import linalg

A = np.array([[2, 0],
              [0, 3]])
eigenvalues, eigenvectors = linalg.eig(A)
print("eigenvalues:", eigenvalues)
```

Fitting a curve to noisy data — a very common real-world task:

```
from scipy.optimize import curve_fit

days = np.arange(7)
# a noisy "warming trend" over the week
rng = np.random.default_rng(seed=0)
noisy_temps = 17 + 0.6 * days + rng.normal(0, 0.5, size=7)

def linear_trend(day, slope, intercept):
    return slope * day + intercept

(slope, intercept), _ = curve_fit(linear_trend, days, noisy_temps)
print(f"fitted trend: {slope:.2f} °C/day, starting at {intercept:.2f} °C")
```

Exercise: use `scipy.linalg.eig` to find the eigenvalues of the matrix `B = np.array([[4, 1], [2, 3]])`.

```
B = np.array([[4, 1],
              [2, 3]])
eigenvalues, eigenvectors = linalg.eig(B)
print(eigenvalues)
```

6.3 Matplotlib — plotting

Matplotlib is the standard plotting library. The `%matplotlib inline` magic command (Jupyter-specific, not plain Python) makes plots appear directly in the notebook.

```
%matplotlib inline
import matplotlib.pyplot as plt
```

```

days = np.arange(7)

plt.plot(days, temperatures, marker="o")
plt.xlabel("Day")
plt.ylabel("Temperature (°C)")
plt.title("A week of temperature readings")
plt.show()

```

A bar chart — useful for comparing categories:

```

subjects = ["Maths", "Physics", "Chemistry", "Biology"]
average_scores = [78, 85, 71, 90]

plt.bar(subjects, average_scores)
plt.xlabel("Subject")
plt.ylabel("Average score")
plt.title("Average score by subject")
plt.show()

```

Multiple series and a legend — e.g. comparing the raw data to the fitted trend from section 6.2:

```

fig, ax = plt.subplots(figsize=(6, 4))
ax.scatter(days, noisy_temps, label="observed")
ax.plot(days, linear_trend(days, slope, intercept), color="orange", label="fitted trend")
ax.set_xlabel("Day")
ax.set_ylabel("Temperature (°C)")
ax.set_title("Observed vs. fitted temperature trend")
ax.legend()
plt.show()

```

(stretch) A scatter plot showing every point from a couple of arrays, with annotations:

```

labels = ["A", "B", "C", "D"]
x_vals = np.array([1, 2, 3, 4])
y_vals = np.array([10, 14, 9, 18])

fig, ax = plt.subplots(figsize=(5, 4))
ax.scatter(x_vals, y_vals)
for label, x, y in zip(labels, x_vals, y_vals):
    ax.annotate(label, (x, y), textcoords="offset points", xytext=(5, 5))
ax.set_xlabel("x")
ax.set_ylabel("y")
ax.set_title("Labelled scatter plot")
plt.show()

```

Exercise: using the `heights` array from the NumPy exercise, make a bar chart with one bar per person, showing their height in centimetres. Label the axes and give the chart a title.

```

people = ["P1", "P2", "P3", "P4", "P5"]
heights_cm = heights * 100

plt.bar(people, heights_cm)
plt.xlabel("Person")
plt.ylabel("Height (cm)")
plt.title("Heights of five people")
plt.show()

```

7. Wrap-up

What you now have:

- ♦ Core Python syntax: variables, types, operators, `if/for/while`
- ♦ The four built-in data structures: `list`, `tuple`, `dict`, `set`
- ♦ Functions, and classes/objects — including composing objects out of other objects, and recognising that library objects (like `datetime.date` or `pathlib.Path`) work exactly the same way as the classes you wrote yourself
- ♦ How to manage packages and environments with Anaconda (`conda`) and `pip`
- ♦ NumPy arrays and linear algebra, a taste of SciPy, and Matplotlib plotting

Common early pitfalls to watch for:

- ♦ Indentation is *meaningful* in Python — a misaligned line under an `if/for/def` is a syntax error, not a style choice.
- ♦ `=` is assignment; `==` is comparison. `if x = 5` is a syntax error (which at least fails loudly).
- ♦ Indexing starts at 0, not 1.
- ♦ Running cells out of order can leave your notebook in a confusing state — if things look wrong, use the “Restart Kernel and Run All” option.
- ♦ `numpy` arrays vs. Python lists: `[1, 2] + [3, 4]` concatenates (`[1, 2, 3, 4]`), but `np.array([1, 2]) + np.array([3, 4])` adds element-wise (`[4, 6]`). Easy to trip over.

Next up: a dedicated session on quantum computing and Qiskit, building directly on the Python, NumPy and Matplotlib skills from today.

Questions? Ask now, or flag a TA during the break.